

B. E. II Examination Nov-Dec 2023
Information Technology
3ITRG1: Computer Organization and Architecture

Duration: 3 Hrs

Max. Marks: 60

Total No. of Questions: 05

Note: All questions are compulsory. Attempt any two parts from each question. Make suitable assumptions if necessary. The answer should be to the point.

- Q1.** A. Give a historical perspective of computers in terms of generations and evolution of performance. **06**
- B. Define the following: **06**
- | | | |
|---------------------|----------------------|----------------|
| 1. Desktop Computer | 2. Server | 3. Data Center |
| 4. Super Computer | 5. Embedded Computer | 6. Petabyte |
- C. Write the following rules for arithmetic operations on floating-point numbers: **06**
- | | | |
|----------------------|------------------|----------------|
| 1. Add/Subtract Rule | 2. Multiply Rule | 3. Divide Rule |
|----------------------|------------------|----------------|
- Q2.** A. Explain caches on the processor chip. **06**
- B. Criticize the following statement: "Using a faster processor chip results in a corresponding increase in performance of a computer even if the main memory speed remains the same." **06**
- C. Consider a computer that has a cache that holds 8-word blocks and a typical DRAM main memory. 17 cycles are needed to load a block into the cache. Assume that 30 percent of the instruction in a typical program performs a read or a write operation. Assume that the hit rate in the cache is 0.95 for instructions and 0.9 for data. 1 clock cycle is needed to access information in the cache. Let us further assume that the miss penalty is the same for both read and write accesses. **06**
- How effective this cache is compared to an ideal cache?
- Q3.** A. For a given CPU, the first machine cycle M1 always consists of 4 states, and the remaining machine cycles consist either of 3 or 4 states. An instruction cycle consists of a minimum of 2 and a maximum of 5 machine cycles. If the CPU's clock rate is 50 MHz and every memory access introduces one wait state. Then calculate: **06**
- A). The shortest instruction execution time,
- B). The longest instruction execution time,
- B. Design a firmware architecture for a digital component that has four states, three inputs (J, K, and Clock), and two outputs (ZERO and ONE). **06**
- C. What are the advantages and disadvantages of hardwired and microprogrammed control? **06**